BLOCKCHAIN INNOVATION CHALLENGE · PROBLEM SPACE 3

OpenTab Global

Family support, spendable at the corner shop, approved on a phone with no data, settled on Stellar.

THE PROBLEM

The money arrives. Then it has to become groceries.

- Remittances are often collected as cash at a pickup counter first.
- Many recipients have a phone but unreliable or costly mobile data.
- Senders can't earmark support for essentials.
- Small shops extend informal credit and carry the risk.

THE SOLUTION

A shared family balance, spendable at local shops

SENDER · ONLINE
Funds
the balance

→

SHOP · ONLINE
Invoice QR
local price + exact charge

→

PHONE · OFFLINE
Signs
that exact purchase

→

SHOP · ONLINE
Submits

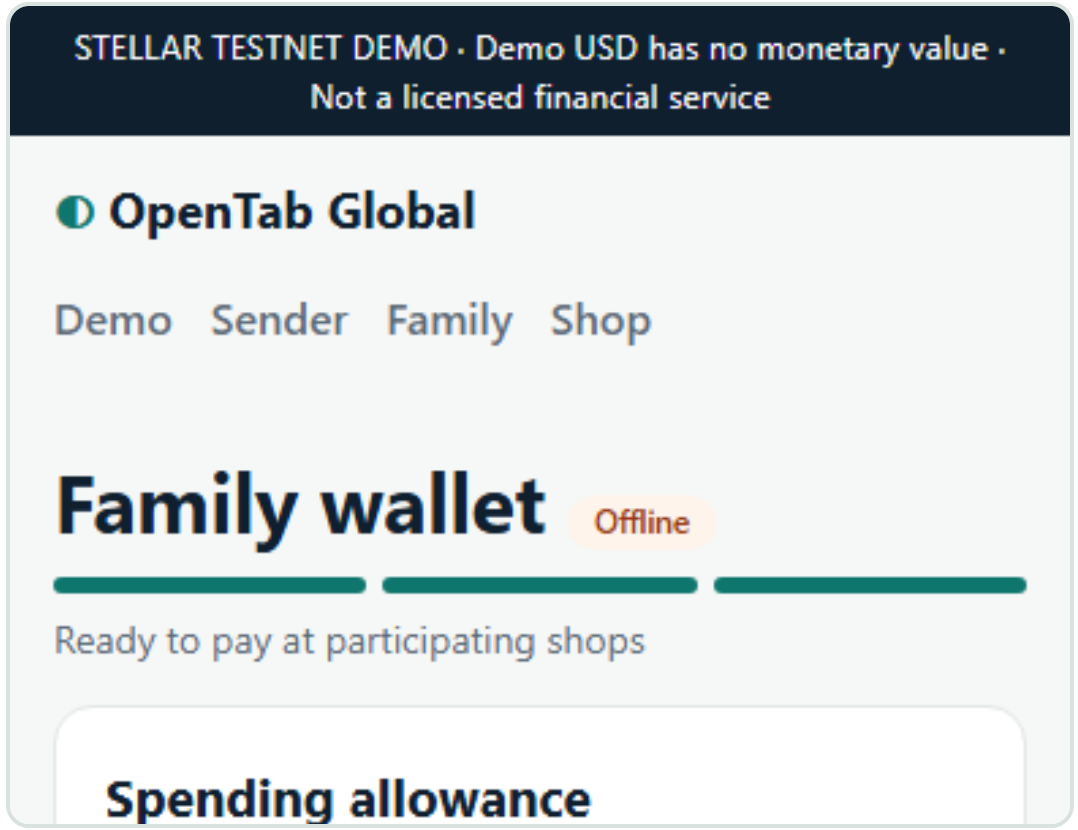
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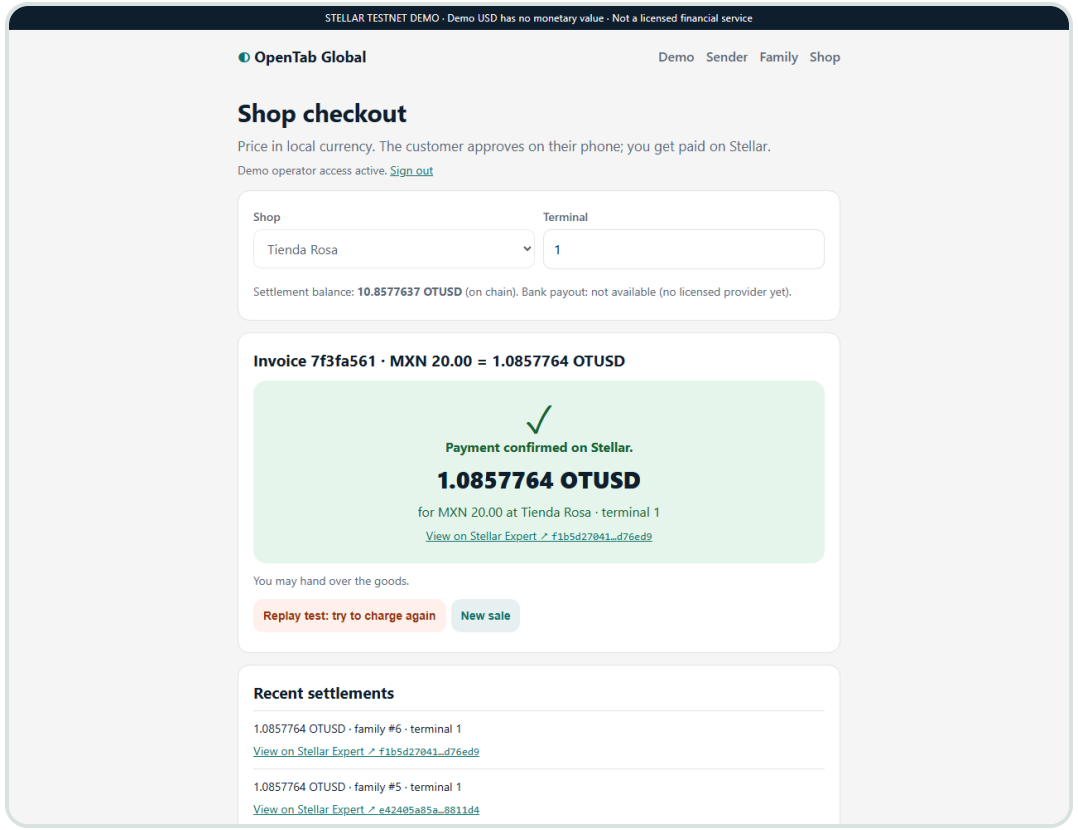
STELLAR
Pays the shop
once

The family member needs **no internet, no crypto account and no XLM**. The shop sees “Payment confirmed” only after Stellar confirms.

LIVE DEMO

Real app, real Testnet transactions





Left: phone offline, "Authorized. Awaiting merchant settlement." Right: shop after on-chain confirmation, with explorer link.

WHY STELLAR

The rules live in a Soroban contract, not our server

- Each approval binds network, contract, token, family, shop, terminal, invoice, amount, price and expiry.
- Each invoice can be paid once; replays are refused.
- One balance across many shops; two shops can never overspend it together.
- Native Ed25519 verification: no account needed for the shopper.
- Stellar Asset Contracts hold and pay a dollar token.
- Low fees, ~5 s finality, anchor ecosystem for future cash-out.

TECHNICAL PROOF

Verified, not mocked

- FamilyVault deployed: `CAUCH3MNJGIYDLOY3OB3YCKRCBWWY3J3HZB5QCHPJXUWXBMK5IABYLGW`
- Confirmed payments at two shops from one family balance; same approval replayed and refused.
- 34 contract tests · 13 protocol tests · 18 live API checks · 18 security checks · real-browser tests, including signing in an emulated offline browser.
- Signing bytes produced by the browser are verified by the Rust contract tests.

Settlement token: OpenTab Demo USD (OTUSD), Testnet only, no value, not USDC.

OPPORTUNITY AND ROADMAP

Shops paid with certainty; support spent on essentials

Business

- Small settlement fee is a possible model (not yet validated).
- Shops replace informal credit with prepaid, verifiable payment.

Next

- Physical-device pilots
- Both-offline mode: funds reserved for one named shop (in development)
- Real USDC, then licensed local payout partners
- Audit

Your support.
Their everyday essentials.

Testnet prototype · demo token has no value · illustrative exchange rates · no bank payouts yet · unaudited.